SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.

2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

*I **T.RUPENDRA/(192311325)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:T.RUPENDRA

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

CO-4 AT-3 class test

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(1) Modran web applications used several technologies to manage data, process requests, communications with database, & present information to users,

XML (Extensible Markup Language)

it used to store, structure, & transport data in a platform independent formate.

Ex:-

<student>
<name> Ravi</nam>
<rollno>101 </rollno>
<Dep> CSE </Dep>
</Student.

* stores structural data
* Transfers data b/w different app
* separates data from

JSP (Java ~~script~~ Server pages)

it is a Java based server side technology used to creat dynamic web pages.

* Generates dynamic webcontent.
* processes user requests.
* Display data

MVC Architecture

MVC (model-view-controller) is a software architectural pattern that divides an application in to three logical components.

flow Chart

user → controller → model → Database

Database → model → controller → view → user

* separates presentation & business logic.

XSLT (Extensible stylesheet Language Transformations)

it is a language used to transform XML documents in to another formats HTML, XML,

XML → XSLT → HTML

XML Schema

it is commonly called XSD (XML Schema Definition). It is used to define and validate the structure & data types of XML documents

JSTL (JSP Standard Tag library)

it provides pre defined tags that simplify common operations

PHP

it is a server side scripting language widely used for developing dynamic web applications.

```
<?php
$name = "Ravi";
echo "welcome". $name;
?>
```

Database Connectivity

it refers to the process of connecting a web application.

parser

A parser is a software component that reads structured data & analyzes it according to a particular syntax or grammar.

Interconnection of these Technologies in a web Application

These technologies work together to provide data storage, validation, processing, presentation, and communication in a web application.

XML -> XML Schema

XML is used to represent structured data. XML schema (XSD) defines the rules that the XML document must follow. The XML document must follow. The XML document is validated against the schema to ensure that its structure & data types are correct.

XML → parser

An xml parser reads the xml document and analyzes its structure. it allows the applica to access elements & values stored in the XML document.

XML → XSLT

After XML data is processed, XSLT can transform the XML into HTML or another XML format so that the data can be displayed to users.

JSP/ PHP → Server-side processing

JSP & PHP process requests on the server. They can recive form data, execute business logic access XML data, communication with database & generate dynamic responses.

MVC → Application organization

MVC provides the overall structure of the Application.

user → controller → model → Database → model → view → user

* The controller receives the user's request.

* The Model performs business logic & handles data

* The Database connectivity layer communicates with the database

*XML parsers can process XML data.

* The view, commonly implemented using JSP or generated HTML, displays the result

* JSTL can be used within JSP to perform common presentation tasks.

Database Connectivity → Data processing

JSP / Java or PHP can use database Connectivity with the database. Data can be retrived processed, updated, or stored based on user requests.

Overall flow

User request → Controller → JSP/PHP → Bussiness logic → Database /HTML → parser → Data processing → XSLT /JSP view → HTML Response → user.

XML & XML Schema mange structred data

parsers process XML XSLT transforms

XML, JSP /PHP perform server-side provides

MVC organizes the application, JSTL simplifies

JSP development, & database connecting

enables persistent data storage & retrivel.

Together, these technologies help create

structred, dynamic, maintainable, & data

-driven web application

XML & XML Schema manage structured data parsers process XML XSLT transforms XML, JSP /PHP perform server-side provides MVC organizes the application, JSTL simplifies JSP development, & database connectivity enables persistent data storage & retrival.

Together, these technologies help create structured, dynamic, maintainable, & data-driven web application.